

Switching Techniques in Routers

Switching Fabric is the core internal subsystem of a router responsible for transferring packets from input ports to output ports. It determines:

- overall forwarding speed,
- degree of internal parallelism,
- scalability of the router.

Routers typically use one of the three switching architectures:

1. Switching via Memory
2. Switching via a Shared Bus
3. Switching via an Interconnection Network (Crossbar, Multistage)

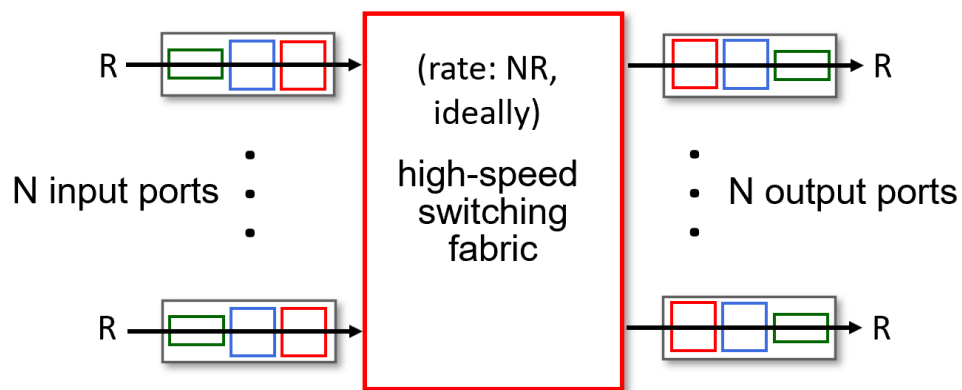


Figure 1: **Switching**

Switching via Memory

In this earliest architecture, the router behaves like a traditional computer: all packet transfers occur through shared system memory.

Packet Flow

Step 1: Input port receives a packet and interrupts the CPU.

Step 2: CPU copies packet into main memory.

Step 3: CPU extracts destination address and performs forwarding-table lookup.

Step 4: CPU copies packet from memory to the appropriate output port buffer.

Performance Limitation

If memory bandwidth is B operations/s, forwarding throughput is:

$$\text{Throughput} = \frac{B}{2}$$

because each packet needs a **read** + **write**.

Thus, ****memory bandwidth becomes the bottleneck****.

Modern Enhancements

- Input line cards now perform lookup and writing operations.
- Reduces CPU load.
- Architecture resembles a shared-memory multiprocessor.
- Example: Cisco Catalyst 8500.

When useful: Low to medium throughput routers where simplicity is preferred.

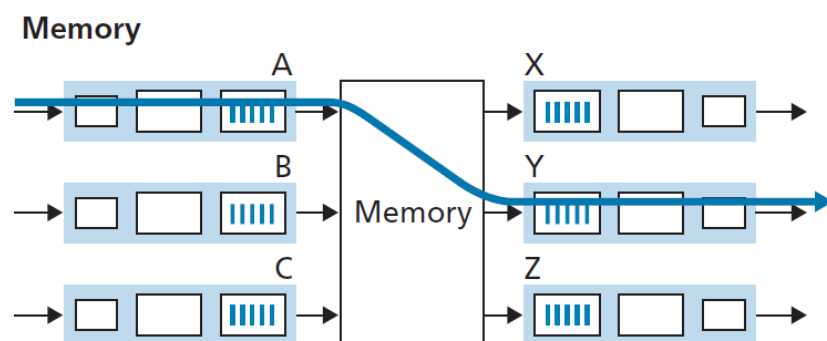


Figure 2: Memory Architecture Diagram

Switching via a Bus

A shared internal bus connects all input ports and output ports.

Operation

- Input port attaches a temporary internal label containing the target output port.
- Packet is broadcast onto the shared bus.
- All output ports receive it, but only the matching one accepts it.
- Label is removed after reception.

Limitation: Bus Contention

Only one packet can traverse the bus at a time. Multiple simultaneous arrivals must queue. Switching speed is limited by the bus bandwidth.

Usage and Example

- Used in LAN/enterprise routers.
- Sufficient for moderate traffic.
- Example: Cisco 6500 with 32 Gbps backplane bus.

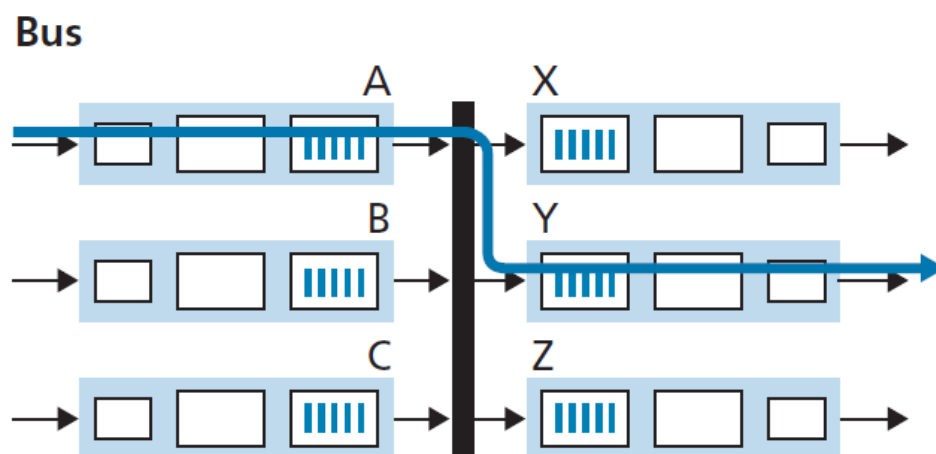


Figure 3: Bus Interconnection Diagram

Switching via an Interconnection Network

Designed to overcome bottlenecks of memory and bus architectures.

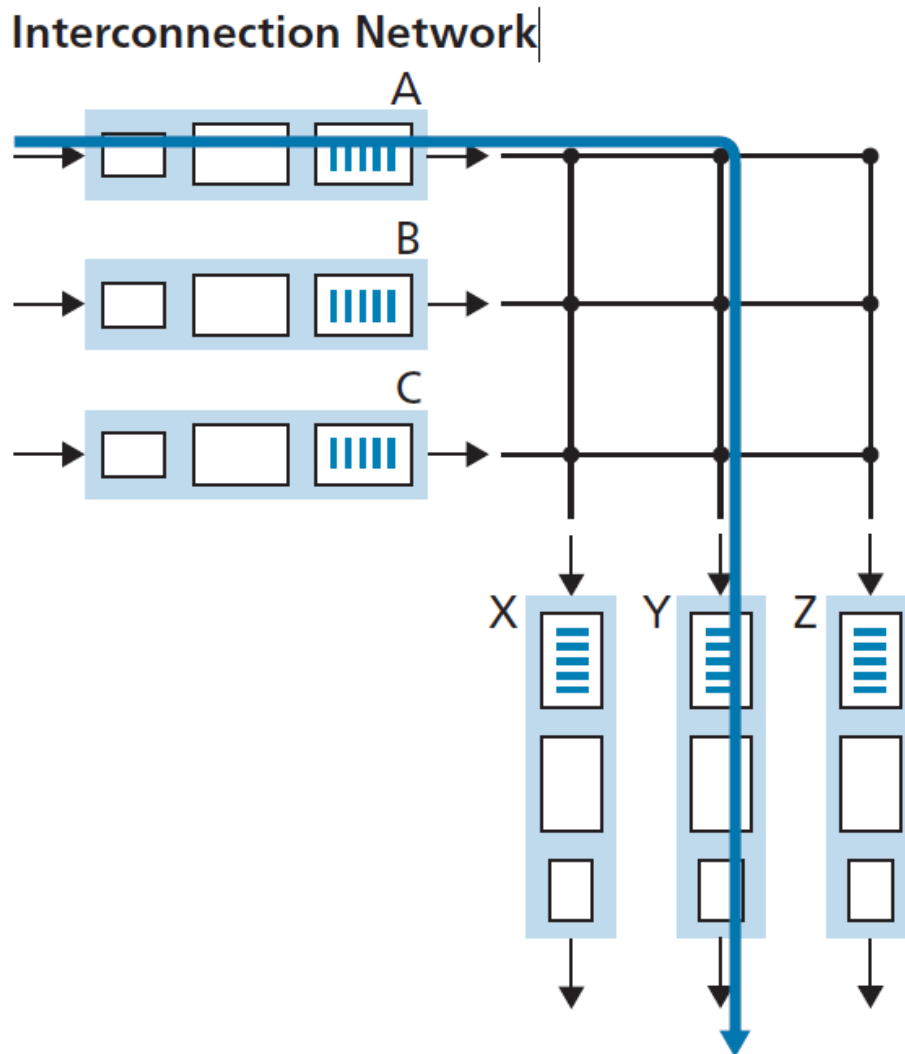


Figure 4: Interconnection Network Architecture

Crossbar Switch

A crossbar consists of N input lines and N output lines connected through $N \times N$ crosspoints.

- Crosspoints open/close under fabric controller.
- Input A can forward to output Y independently of other pairs.
- Supports true parallelism.

Parallel Forwarding

A crossbar is **non-blocking** as long as multiple packets do not target the same output port.

Examples

- Cisco 12000 (crossbar)
- Cisco 7600 (bus or crossbar)

Characteristics

- High throughput
- Parallel forwarding
- More costly and complex than a bus

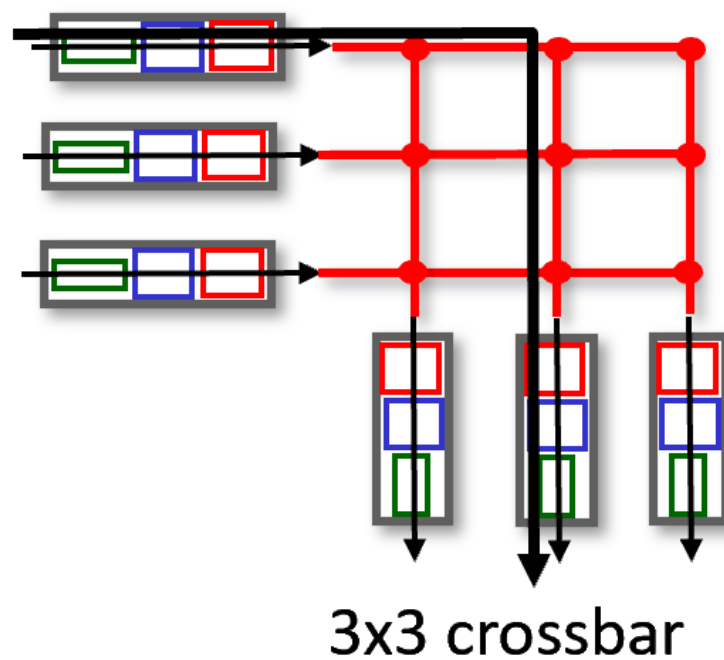


Figure 5: Crossbar Switch Architecture

Multistage Switching Fabrics

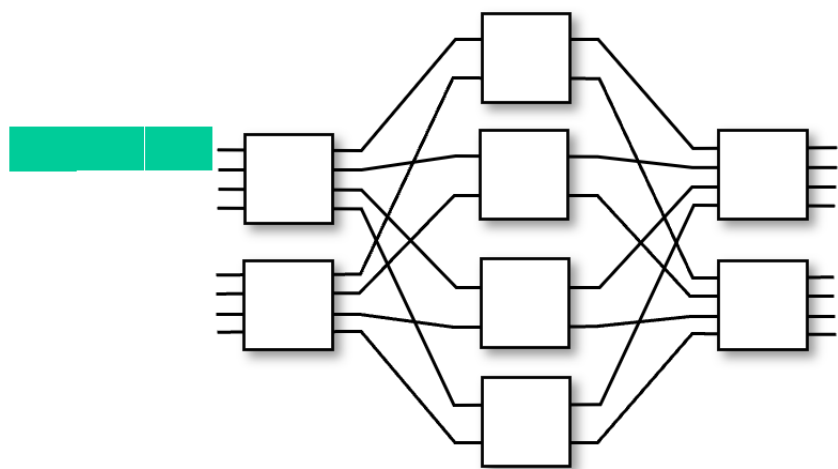
Large routers use multi-layer switching networks instead of a single crossbar.

Why Multistage?

- Single crossbar scales poorly for large N
- Multistage architectures reduce hardware complexity
- Provide higher aggregate bandwidth

Real Example

- Cisco CRS uses a **three-stage, non-blocking** architecture.



8x8 multistage switch
built from smaller-sized switches

Network Layer

Figure 6: Multistage Switching Network

Parallel Switching Fabrics

High-end routers may run several switching fabrics in parallel.

How It Works

Step 1: Input port splits packet into K equal chunks.

Step 2: Chunks are distributed across N parallel fabrics.

Step 3: Output port reassembles the chunks.

Benefits

- Dramatically increases switching capacity.
- Scales upwards by adding more fabrics.

Used in ultra high-capacity core routers where per-slot throughput exceeds a single fabric's capabilities.

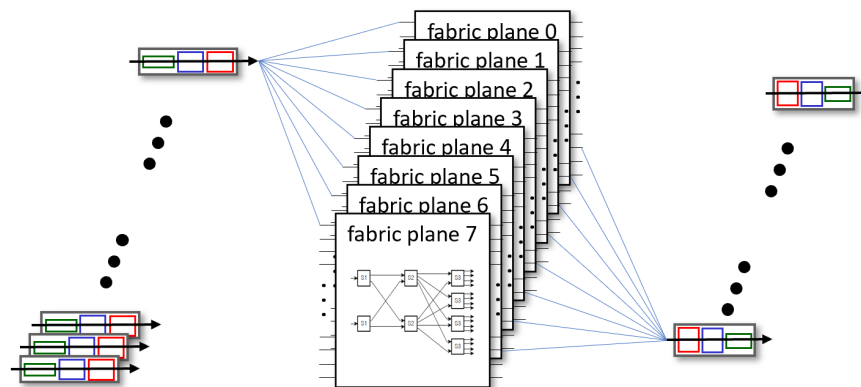


Figure 7: **Parallel Planes Interconnection Architecture**

Summary Comparison

Technique	Parallelism	Bottleneck	Scalability
Memory	None	Memory Bandwidth	Low
Bus	None	Bus Speed	Low–Medium
Crossbar	Yes	Output Port Conflicts	High
Multistage	Yes	Stage Congestion	Very High
Parallel Fabrics	Yes	Reassembly Overhead	Extremely High